Lab Report 1(a):

To write this report, you can use proteus or hardware instruments taken from the lab (remember, if you take hardware instruments from the lab, then return it on the same day, otherwise the lab won't give you the instruments, may blacklist you. Room no. 507 is free for students to conduct experiments). If you use proteus, take screenshots of the circuits you prepared in the proteus and use it in the lab report. If you use hardwares from the lab, then take a clear picture of the overall circuit and then use it in your lab report.

Finally, you must submit the codes of each problem in the lab report. Check Lab reports samples/guidelines document to properly write your lab report. Maintain the formatting of the lab report as mentioned in the documents.

The tasks for lab report 1(a):

1. Use a push button to toggle an LED.
2. Use a push button to turn ON a buzzer.
3. **LDR** sensors can sense the change of intensity of light.Suppose,you have an LDR sensor and an LED . Interface an LDR sensor with Arduino in Proteus and turn on the LED when the intensity of light is low and turn it off when the intensity is high. Use the link given in references to understand how the LDR sensor works. Write in short on how an LDR sensor works and show the output results of this particular problem when light is low or high.
4. Imagine you are doing a project on obstacle avoidance.You want to measure the distance of an object from your device. Interface a **SONAR s**ensor with Arduino and measure distance using it.Show the distance in meter.Use the link given in references to understand how sonar sensor works. Also, download the ultrasonic library for proteus. Also, write in short on how an ultrasonic sensor works.